5 if x then y else succ(zero)

1 x   2 y   4 succ(zero)

3 zero

3) $W(\text{zero}) \rightsquigarrow \emptyset \vdash \text{zero} : Nat$

2) $W(y) \rightsquigarrow y : Y_1 \vdash y : Y_1$

1) $W(x) \rightsquigarrow x_1 : X_1 \vdash x_1 : X_1$

4) $W(\text{succ}(\text{zero})) \rightsquigarrow \vdash \text{succ}(\text{zero}) : Nat$

$mgu(\{Nat \doteq Nat\}) = \emptyset$

5) $W(\text{if } x \text{ then } y \text{ else succ}(\text{zero}))$ (+)

$mgu(\{Y_1 \doteq Nat,\ X_1 \doteq Bool\}) \longrightarrow \text{elim } \{Y_1 := Nat\} \Rightarrow$

$\{X_1 \doteq Bool\} \longrightarrow \text{elim } \{X_1 := Bool\}$

$\emptyset$

$= \{Y_1 := Nat,\ X_1 := Bool\}$

(+) $\rightsquigarrow X_1 : Bool,\ Y_1 : Nat \vdash \text{if } x \text{ then } y \text{ else succ}(\text{zero}) : Nat$

6) $W(\lambda y.\ \text{if } x \text{ then } y \text{ else succ}(\text{zero}))$ (+)

(+) $\rightsquigarrow X_1 : Bool \vdash \lambda y : Nat.\ \text{if } x \text{ then } y \text{ else succ}(\text{zero}) : Nat \to Nat$

7) $W(\lambda x.\ \lambda y.\ \text{if } x \text{ then } y \text{ else succ}(\text{zero}))$

$\rightsquigarrow \vdash \lambda x : Bool.\ \lambda y : Nat.\ \text{if } x \text{ then } y \text{ else succ}(\text{zero}) : Bool \to (Nat \to Nat)$